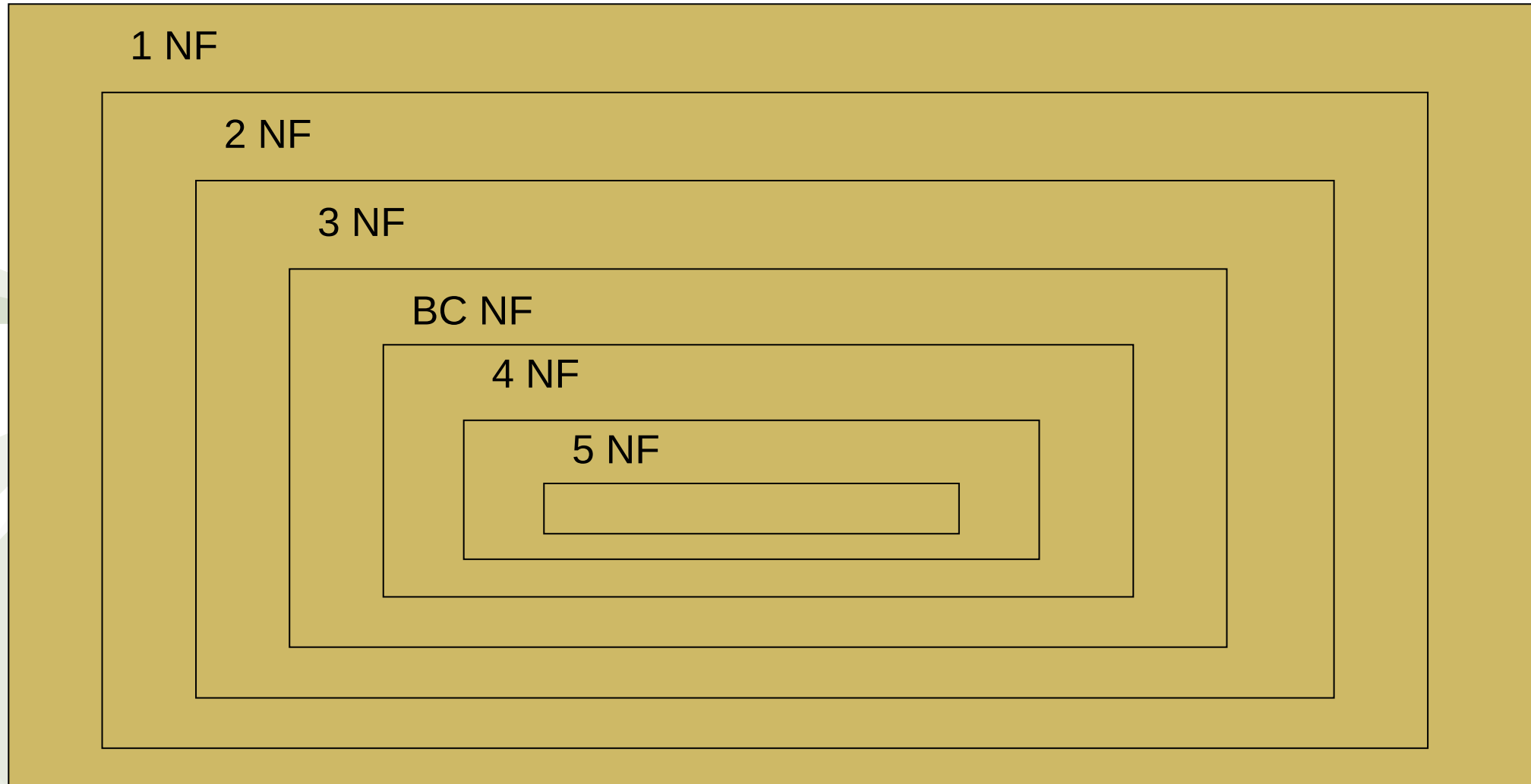


Normalization

Levels of Normalization



Non Loss Decomposition

S

S#	STATUS	CITY
S3	30	Paris
S4	30	Athens

(a) SST

S#	STATUS
S3	30
S4	30

SC

S#	CITY
S3	Paris
S4	Athens

(b) SST

S#	STATUS
S3	30
S4	30

STC

STATUS	CITY
30	Paris
30	Athens

First Normal Form (1NF)

- 1NF Definition :

A relation is in 1NF iff, every tuple contains exactly one value for each attribute.

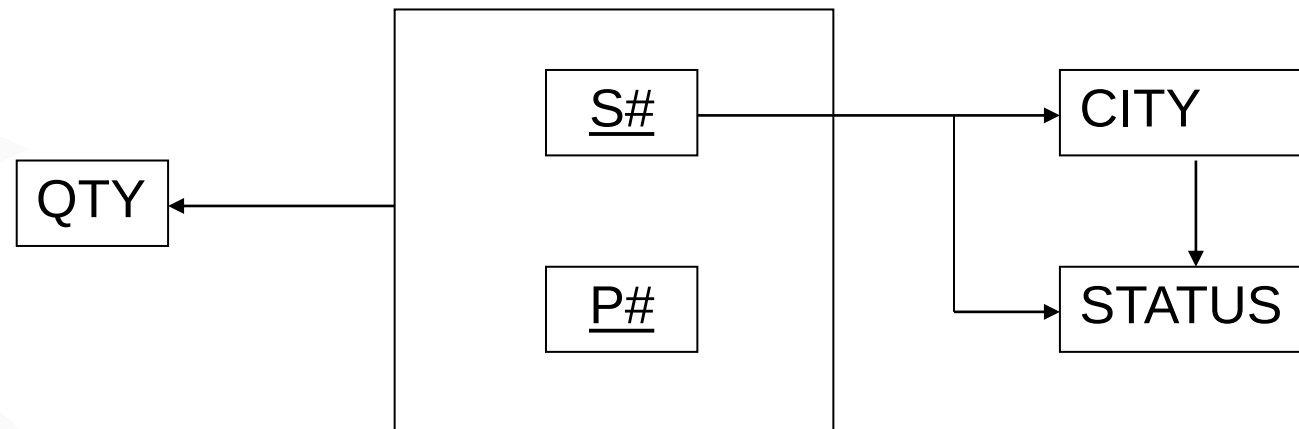
[The only attribute values permitted by 1NF are single atomic values (indivisible/irreducible) values]

1NF (cont.)

Example

FIRST { S#, STATUS, CITY, P#, QTY }

Primary Key { S#, P# }



FD Diagram of Relation FIRST

1NF (cont.)

FIRST	<u>S#</u>	STATUS	CITY	<u>P#</u>	QTY
	S1	20	London	P1	300
	S1	20	London	P2	200
	S1	20	London	P3	400
	S1	20	London	P4	200
	S1	20	London	P5	100
	S1	20	London	P6	100
	S2	10	Paris	P1	300
	S2	10	Paris	P2	400
	S3	10	Paris	P2	200
	S4	20	London	P2	200
	S4	20	London	P4	300
	S4	20	London	P5	400

Problems in any NF

- Data Redundancy in Tuples
- Update Anomalies (reasons from data redundancy)
 - Insert : can not insert the fact that a particular supplier is located in a particular city until that supplier supplies at least one part (ex. supplier S5 is located in Athens)

Problems in any NF (cont.)

- Delete : if we delete the sole FIRST tuple for a particular supplier, we delete not only the shipment connecting that supplier to a particular part but also the information that the supplier is located in a particular city.

(ex. if delete tuple $S\# = S3$ and $P\# = P2$,
will lose information $S3$ is located in Paris)

Problems in any NF (cont.)

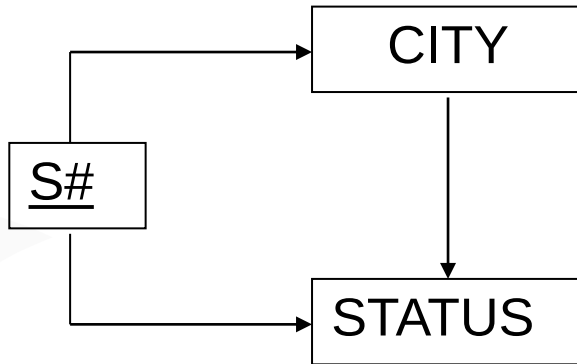
- Update : the city for a given supplier appears in FIRST many times.

this redundancy cause update problem

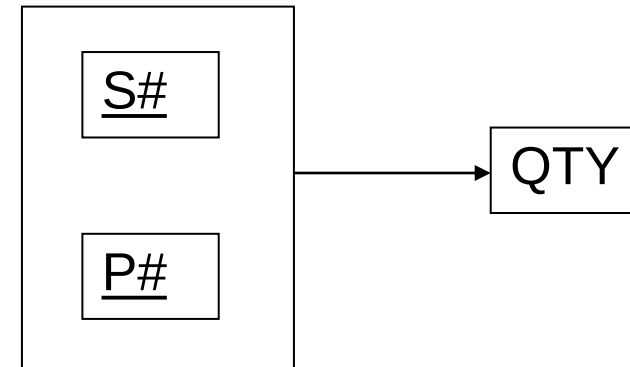
(ex. If supplier S1 move to Amsterdam, the problem to search of S1 and change data from London to Amsterdam, may be cause of inconsistent result)

2NF

- The solution of 1NF problems by separate the FIRST relation into 2 relations



SECOND Relation



SP Relation

2NF (cont.)

SECOND

<u>S#</u>	STATUS	CITY
S1	20	London
S2	10	Paris
S3	10	Paris
S4	20	London
S5	30	Athens

SP

<u>S#</u>	<u>P#</u>	QTY
S1	P1	300
S1	P2	200
S1	P3	400
S1	P4	200
S1	P5	100
S1	P6	100
S2	P1	300
S2	P2	400
S3	P2	200
S4	P2	200
S4	P4	300
S4	P5	400

2NF (cont.)

- 2NF Definition :

A relation is in 2NF iff it is in 1NF and every nonkey attribute is irreducibly dependent on the primary key.

[2NF is based on the concept of full functional dependency (not partial dependency) on the PK]

- 2NF update anomalies problems?

2NF (cont.)

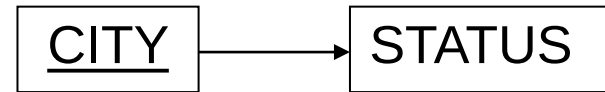
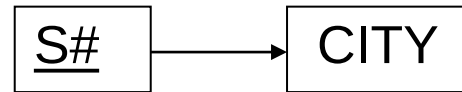
- Insert : cannot insert the fact that a particular city has a particular status (e.g. Rome has a status 50, until we have some supplier actually located in that city)
- Delete : if delete the sole SECOND tuple for a particular city the information of supplier , sole and status data (e.g. if we delete S5, we lose the information that the status for Athens is 30)

2NF (cont.)

- Update : the status for a given city appears in SECOND many times (e.g. to change the status of London from 20 to 30)

3NF

- The solution of 2NF problems by separate the SECOND relation into 2 relations



FDs for relation SC and CS

SC

<u>S#</u>	CITY
S1	London
S2	Paris
S3	Paris
S4	London
S5	Athens

CS

<u>CITY</u>	STATUS
Athens	30
London	20
Paris	10
Romes	50

3NF (cont.)

- 3NF Definition :
A relation is in 3NF iff it is in 2NF and every nonkey attribute is nontransitively dependent on the PK. (3NF is based on the concept of transitive dependency)
- Nonloss decomposition problem?
(a) SC {S#, CITY} and CS {CITY, STATUS} or
(b) SC {S#, CITY} and SS {S#, STATUS}
- It is still not possible in (b) to insert the information that a particular city has a particular status unless some supplier is located in that city

Boyce/Codd Normal form (BCNF)

- Formal BCNF Definition : A Relation is in BCNF iff every nontrivial, left irreducible FD has a candidate key as its determinant
- Informal BCNF Definition : A relation is in BCNF iff the determinants are candidate key
(every relation in BCNF is also in 3NF, however, a relation in 3NF is not necessarily in BCNF)

BCNF (cont.)

SJT

<u>S</u>	<u>J</u>	T
Smith	Math	Prof. White
Smith	Physics	Prof. green
Jones	Math	Prof. White
Jones	Physics	Prof. Brown

```
graph LR; subgraph Box; S[S]; J[J]; end; T[T]; S --> T; J --> T;
```

SJT is in 3NF, and not in BCNF, why?

ST { S, T } and TJ { T, J }

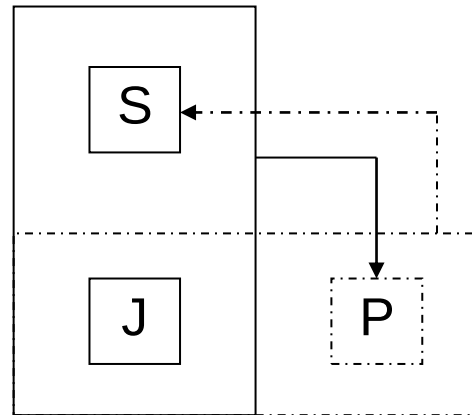
PK in ST and TJ? Are ST and TJ in BCNF?

Decomposition problems?

BCNF (cont.)

- Ex. In EXAM relation, student S was examined in subject J and achieved position P in the class list. There are no two students obtained the same position in the same subject.

EXAM



Is EXAM in BCNF?

4NF

CTX

<u>COURSE</u>	<u>TEACHER</u>	<u>TEXT</u>
Physics	Prof. Green	Basic Mechanics
Physics	Prof. Green	Principles of Optics
Physics	Prof. Brown	Basic Mechanics
Physics	Prof. Brown	Principles of Optics
Math	Prof. Green	Basic Mechanics
Math	Prof. Green	Vector Analysis
Math	Prof. Green	Trigonometry

What is the highest NF that CTX hold?

Redundancy data appear in CTX.

CTX has update anomalies problems?

How to add information? e.g. physics course can be taught by a new teacher, it is necessary to insert two tuples, one for each of the two texts

4NF (cont.)

CT

<u>COURSE</u>	<u>TEACHER</u>
Physics	Prof. Green
Physics	Prof. Brown
Math	Prof. Green

CX

<u>COURSE</u>	<u>TEXT</u>
Physics	Basics Mechanics
Physics	Principles of Optics
Math	Basics Mechanics
Math	Vector Analysis
Math	Trigonometry

Now we can insert a single tuple in each relation, and

COURSE is not **multi-valued determine** on TEACHER and TEXT

4NF (cont.)

- Multi-valued dependence : Let R be a relation, and let A , B , and C be subsets of the attributes of R . Then we say that B is multi-dependent on A , in symbols

$$A \twoheadrightarrow B \mid C \text{ (A multi-determines B or C)}$$

iff in every possible legal value of R , the set of R values matching a given (A value or C value) pair depends only on the A value and is independent of the C value

4NF (cont.)

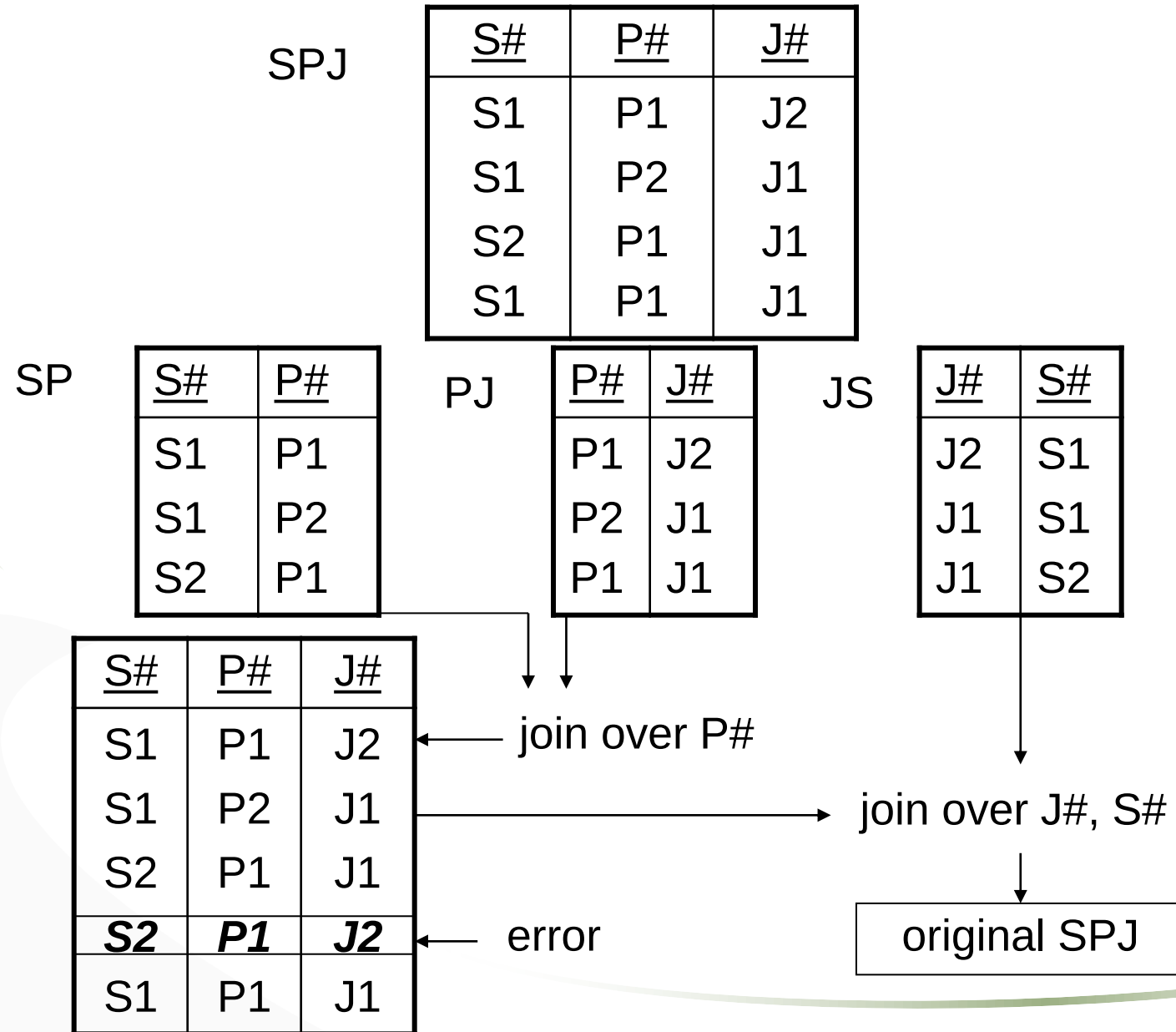
- Ex. $\text{COURSE} \twoheadrightarrow \twoheadrightarrow \text{TEACHER} \mid \text{TEXT}$
- MVD Theorem (Fagin) : Let $R\{A, B, C\}$ be a relation, where A , B , and C are sets of attributes. Then R is equal to the join of its projection on $\{A, B\}$ and $\{A, C\}$ iff R satisfies the MVDs $A \twoheadrightarrow \twoheadrightarrow B \mid C$
- An FD is **trivial** iff the right-hand side is a subset of the left-hand side e.g.

$$\{ S\#, P\# \} \rightarrow S\#$$

4NF (cont.)

- 4NF definition :
- Relation R is in 4NF iff, whenever there exist subsets A and B of the attributes of R such that the nontrivial MVD $A \twoheadrightarrow B$ is satisfied, then all attributes of R are also functionally dependent on A .
(R is in 4NF iff it is in BCNF and there are no MVDs)

5NF



5NF (cont.)

- Relation SPJ is the join of all three of its binary projections but not any two called Constraint 3D (3-decomposable)
- Constraint 3D is satisfied iff the relation concerned is equal to the join of certain of its projections, it constraint as a join dependency (JD)

5NF (cont.)

- Join dependency : Let R be a relation, and let $A, B, \dots Z$ be subsets of the attributes of R . Then R satisfies the JD with symbol
* $\{A, B, \dots, Z\}$ or star $A, B, \dots Z$
iff every possible legal value of R is equal to the join of its projections on $A, B, \dots Z$
- Relation SPJ satisfies the
 $JD * \{SP, PJ, JS\}$

5NF (cont.)

- 5NF definition :

A relation R is in 5NF (also called projection-join normal form or PJ/NF) iff every nontrivial join dependency that hold for R is implied by the candidate key of R

(relation is in 5NF iff there are no join dependency exist)

5NF (cont.)

- Ex. Building-material-supplier

<u>BuildingID</u>	<u>Material</u>	<u>SupplierNo</u>
B45	Sheet Rock	S67
B45	Ceiling Paint	S72
B51	Sheet Rock	S72
B51	Shower Stall	S67
B93	Ceiling Paint	S75

Building B45 gets sheet rock from supplier S67 and ceiling paint from supplier S72. You have a constraint that suppliers may supply only certain materials to specific buildings even though a

supplier may be able to supply all materials. Ex. Supplier S72 can supply sheet rock to building B45, but to this building B45, only supplier S67 is designated to supply sheet rock. This constraint imposes a join dependency on the relation.

5NF (cont.)

- The solution to remove join dependencies

BM	<u>BuildingID</u>	<u>Material</u>	MS	<u>Material</u>	<u>SupplierNo</u>
	B45	Sheet Rock		Sheet Rock	S67
	B45	Ceiling Paint		Ceiling Paint	S72
	B51	Sheet Rock		Sheet Rock	S72
	B51	Shower Stall		Shower Stall	S67
	B93	Ceiling Paint		Ceiling Paint	S75

BS	<u>BulidingID</u>	<u>SupplierNo</u>
	B45	S67
	B45	S72
	B51	S72
	B51	S67
	B93	S75

5NF (cont.)

- Ex. SPJ relation

S#	P#	J#
S1	P1	J2
S1	P2	J1

- If (S1, P2, J1) is inserted, (S1,P1,J1) must also be inserted

<u>S#</u>	<u>P#</u>	<u>J#</u>
S1	P1	J2
S1	P2	J1
S2	P1	J1
S1	P1	J1

- Can delete (S2, P1, J1) without side effect
- If (S1, P1, J1) deleted, another tuple must also be deleted